

Regularized Bradley–Terry Ratings for NFL Pass Protection and Pass Rush

Maximilian Gebauer¹, Victoria Lee¹, Kenny Watts¹, and Jonathan Pipping-Gamón¹

¹University of Pennsylvania

March 29, 2026

Abstract

Quantifying pass protection and pass rushing at the player level remains a core challenge in football analytics. Box-score outcomes such as sacks and hits contain useful signal, but they are sparse, context-dependent, and typically unadjusted for opponent strength. Using one full NFL season of Hudl tracking-derived blocker–rusher interactions, we build a Bradley–Terry (BT) rating pipeline for trench play. The pipeline estimates two separate ridge-regularized paired-comparison models: (i) a binary win/loss model based on the 2.5-second win definition used in pass block win rate settings, and (ii) a multinomial severity model over `loss/win/hit/sack`. Both models are validated out-of-sample against task-specific baselines and are accompanied by end-to-end and weekly path uncertainty estimates. On a chronological holdout ($n = 32,011$), BT improves win-model log loss versus a global baseline (+0.00676) and a matchup baseline (+0.00137), and improves severity multiclass log loss (+0.00752) over class-frequency baseline. Bootstrap intervals for all primary improvements exclude zero. We report role-specific leaderboards and cumulative weekly uncertainty bands to support both end-of-season and longitudinal interpretation.

1 Introduction

Pass protection and pass rushing are central to modern NFL efficiency and roster construction, but player-level evaluation in the trenches remains difficult. For rushers, box-score outcomes such as sacks and hits are rare and heavily mediated by context, including quarterback time-to-throw, coverage, and game situation. For blockers, the inverse problem holds: a lineman can execute consistently without generating a direct box-score statistic. Aggregate summaries often blur individual ability with opponent quality, help structure, or team environment.

To better isolate line play, ESPN introduced pass block win rate (PBWR), which measures whether a blocker sustains a block for 2.5 seconds on a pass play (Burke, 2018). More recent

tracking-based work has moved beyond sacks alone by modeling how pressure develops in space and time. For example, STRAIN summarizes how quickly defenders close space to the quarterback over time (Nguyen et al., 2023). These approaches provide richer information than sacks alone, but they are usually reported as season-level summaries and do not directly deliver opponent-adjusted player ratings for blockers and rushers in one common framework.

Our goal is to evaluate offensive and defensive line play within one opponent-adjusted paired-comparison framework. We treat each blocker–rusher engagement as a matchup between a pass protector and a pass rusher, then fit regularized Bradley–Terry models (Bradley and Terry, 1952; Friedman et al., 2010; Glickman and Jones, 2025). Unlike the earlier Elo-oriented version of this project, the current paper centers on two separate ridge-regularized BT models: one for the binary 2.5-second win/loss task and one for an outcome-weighted severity task. This separation lets us validate each task against its own baseline and quantify uncertainty in a way that matches the estimand.

Contributions. Our main contributions are:

1. A paired-comparison framework that evaluates blockers and rushers jointly while preserving role-specific interpretation.
2. Separate regularized BT models for binary win/loss and multi-outcome severity.
3. Chronological out-of-sample validation against task-specific baselines, including bootstrap uncertainty for predictive performance.
4. End-of-season leaderboards and cumulative weekly path uncertainty for longitudinal interpretation.

2 Data

We use NFL player-tracking data from the 2021 season (regular season and postseason) provided by Hudl. Tracking coordinates are recorded at 10 Hz and are accompanied by event annotations marking events such as blocking engagements, pass attempts, and sacks.

We restrict attention to dropbacks by retaining plays containing either a forward pass event or a quarterback sack event. Within each retained play, we keep offensive linemen, identified pass rushers, and the quarterback. For each frame t , we compute the Euclidean distance between the quarterback and any non-quarterback player:

$$d_{p,t} = \sqrt{(x_{p,t} - x_{QB,t})^2 + (y_{p,t} - y_{QB,t})^2}. \quad (1)$$

Our unit of analysis is a blocker–rusher interaction, defined from the engagement labels in the tracking data. We generate a binary double-team indicator and set it to 1 when multiple blockers are assigned to the same rusher within overlapping time windows.

After filtering and labeling, the modeling table contains 160,054 blocker–rusher interactions across 34,913 pass plays in 279 games, with 623 rushers and 348 blockers. The double-team rate is 42.68%.

3 Outcomes

We model two targets:

1. **Win/Loss target** (`win_target`): indicator of whether the rusher wins within the 2.5-second definition.
2. **Severity target** (`severity_outcome`): multinomial outcome in `{loss, win, hit, sack}`, with scalar severity mapping 0, 0.2, 0.4, 1.0 for weighted evaluation.

We assign one outcome per interaction using severity priority `sack > hit > win > loss`. A `win` occurs if the rusher gains a decisive advantage within 2.5 seconds, operationalized as becoming closer to the quarterback than the blocker before the 2.5-second threshold. A `sack` is taken directly from event annotations. A `hit` is recorded when the quarterback is contacted behind the line of scrimmage before pass release without a sack. If none of these events occurs, the interaction is labeled `loss` from the rusher’s perspective.

Observed frequencies in the full table are:

$$\mathbb{P}(\text{loss}) = 0.731, \quad \mathbb{P}(\text{win}) = 0.253, \quad \mathbb{P}(\text{hit}) = 0.0092, \quad \mathbb{P}(\text{sack}) = 0.0064.$$

4 Modeling Framework

4.1 Win/Loss Ridge BT

For interaction t , with rusher $i(t)$, blocker $j(t)$, and double-team indicator D_t , the binary BT model is

$$\text{logit } \mathbb{P}(Y_t = 1) = r_{i(t)} - b_{j(t)} + \delta D_t,$$

where $Y_t = 1$ indicates rusher win.

We estimate parameters via ridge-penalized logistic regression:

$$\arg \min_{\theta} \left\{ -\ell(\theta) + \lambda \|\theta\|_2^2 \right\},$$

with λ selected by cross-validation on a log-spaced grid.

Throughout the paper, we report raw BT scores rather than an affine Elo-style rescaling so that figures and tables stay directly tied to the fitted model coefficients.

4.2 Severity Ridge BT

For severity classes $c \in \{\text{loss}, \text{win}, \text{hit}, \text{sack}\}$, we fit a ridge-regularized multinomial BT model:

$$\mathbb{P}(C_t = c) = \frac{\exp(\eta_{t,c})}{\sum_{c'} \exp(\eta_{t,c'})}, \quad \eta_{t,c} = r_{i(t),c} - b_{j(t),c} + \delta_c D_t.$$

The scalar severity prediction is the expected score under class probabilities, using the mapping `loss` = 0, `win` = 0.2, `hit` = 0.4, and `sack` = 1.0.

4.3 Validation Baselines

We use a chronological 80/20 split (train = 128,043, test = 32,011).

Win model baselines:

1. **Global mean baseline**: train-set rusher win proportion.
2. **Matchup baseline (logit mean)**: smoothed rusher/blocker components combined on the logit scale with prior strength 25.

Severity model baselines:

1. **Scalar baseline**: train-set expected severity score.
2. **Multiclass baseline**: train-set marginal class probabilities.

4.4 Uncertainty

We report two complementary uncertainty procedures (Efron and Tibshirani, 1994):

1. **End-to-end bootstrap** ($B = 1000$): resample games, refit with fixed λ selected from full-data CV, and recompute validation metrics and player ratings.
2. **Weekly path bootstrap** ($B = 100$): for each cumulative week checkpoint, resample past games and refit to obtain path-level uncertainty bands.

5 Results

5.1 Model Fit and Validation

Cross-validated ridge selected $\lambda_{\min} = 9.26 \times 10^{-5}$ for win BT and $\lambda_{\min} = 6.53 \times 10^{-5}$ for severity BT. The win BT fit retains 972 nonzero terms at $\lambda_{\min}$.

Table 1 summarizes holdout performance and bootstrap uncertainty. All primary improvements have positive 95% bootstrap intervals.

Table 1: Out-of-sample validation metrics for BT models and baselines. Improvements are baseline minus model (higher is better).

Task	Metric	Model	Baseline	Improvement	95% bootstrap CI
Win vs global	Brier	0.18563	0.18820	0.00257	[0.00230, 0.00431]
Win vs global	Log loss	0.55717	0.56393	0.00676	[0.00598, 0.01117]
Win vs matchup	Brier	0.18563	0.18613	0.00050	[0.00010, 0.00098]
Win vs matchup	Log loss	0.55717	0.55855	0.00137	[0.00025, 0.00251]
Severity (scalar)	MSE	0.01351	0.01361	0.00010	[0.00010, 0.00023]
Severity (multiclass)	Log loss	0.62723	0.63475	0.00752	[0.00790, 0.01590]

5.2 Ratings and Leaderboards

Figure 1 shows raw BT score distributions by role for both tasks. Figure 2 reports top players by model and role (minimum 200 interactions to reduce small-sample volatility). Scores are interpreted within each model (win vs severity) rather than across models, because the two targets induce different scales. The win/loss and severity leaderboards are directionally similar for elite rushers but show clear differences for blockers, which is expected because the severity model upweights high-impact outcomes.

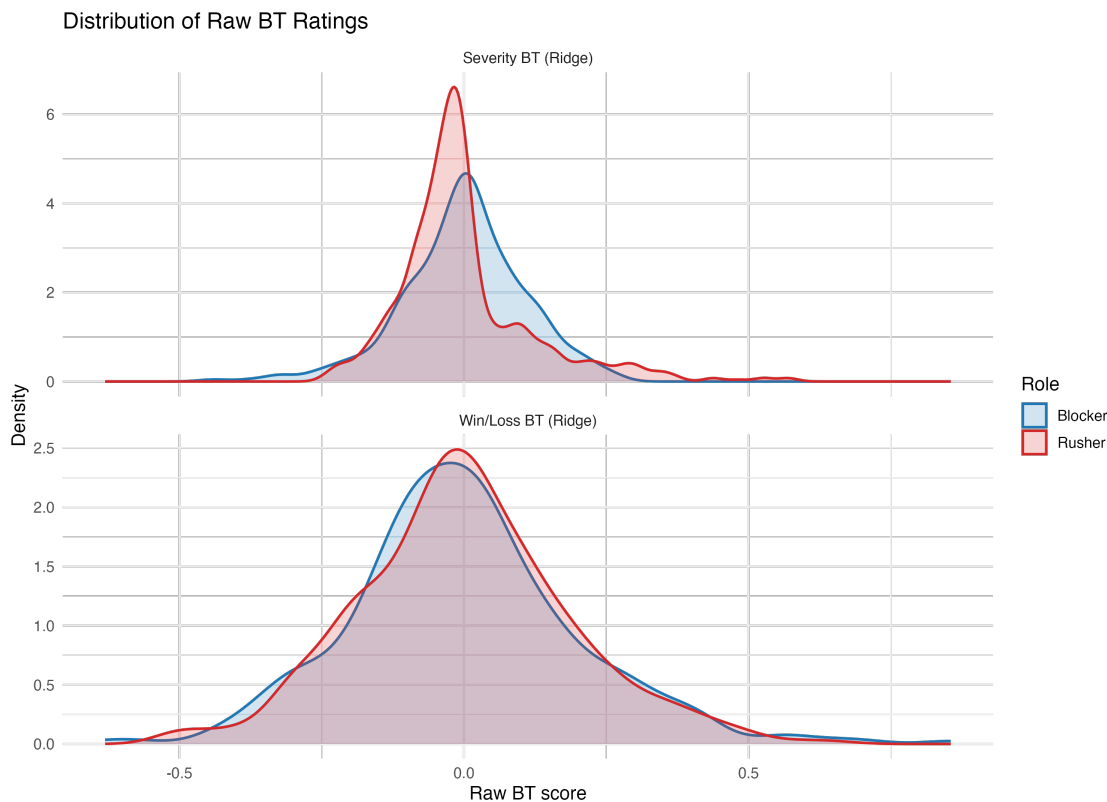


Figure 1: Distribution of raw BT scores by model and role.

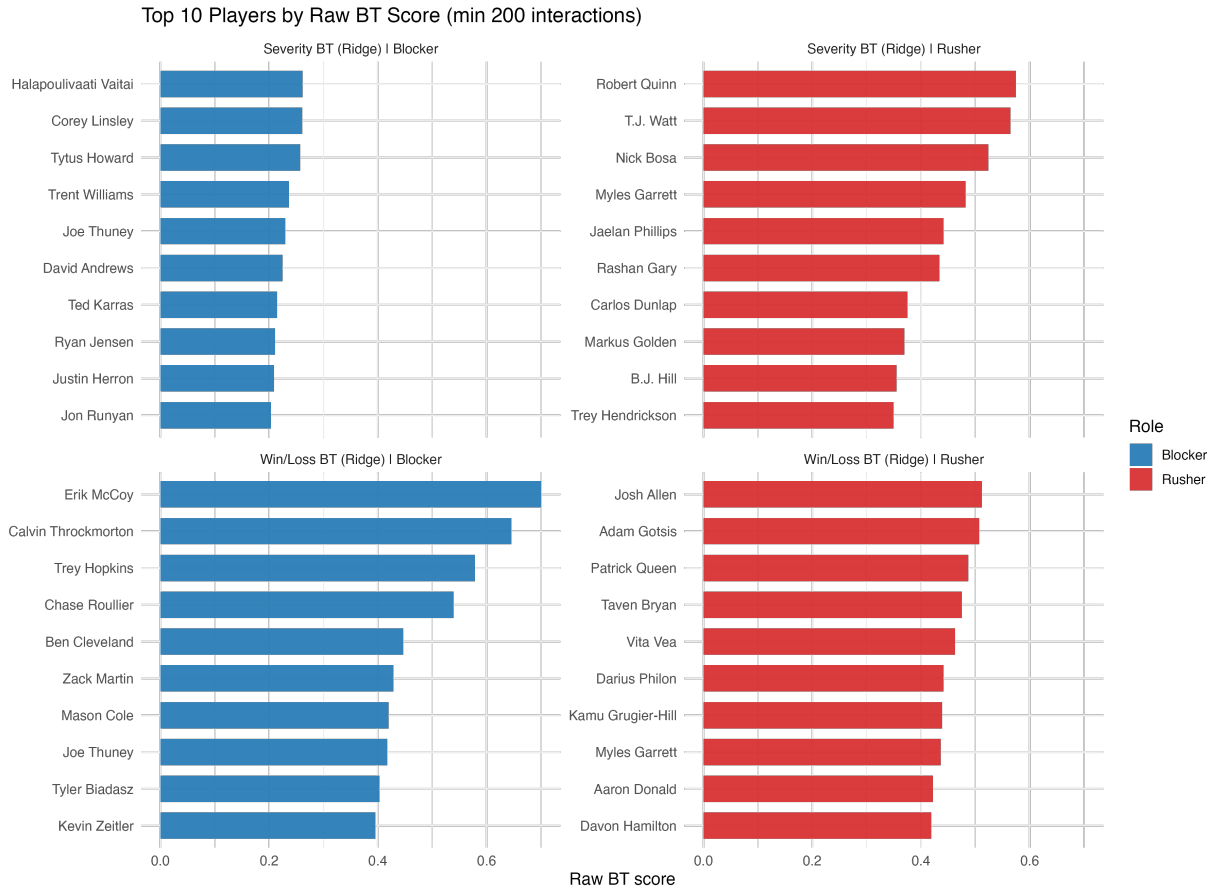


Figure 2: Top 10 players by raw BT score in each model-role panel (minimum 200 interactions).

Table 2: Top five players by model and role (minimum 200 interactions). Values are raw BT scores.

Model	Role	Player	Rating
Win/Loss BT	Rusher	Josh Allen	0.512
Win/Loss BT	Rusher	Adam Gotsis	0.507
Win/Loss BT	Rusher	Patrick Queen	0.487
Win/Loss BT	Rusher	Taven Bryan	0.475
Win/Loss BT	Rusher	Vita Vea	0.463
Win/Loss BT	Blocker	Erik McCoy	0.700
Win/Loss BT	Blocker	Calvin Throckmorton	0.646
Win/Loss BT	Blocker	Trey Hopkins	0.579
Win/Loss BT	Blocker	Chase Roullier	0.539
Win/Loss BT	Blocker	Ben Cleveland	0.447
Severity BT	Rusher	Robert Quinn	0.575
Severity BT	Rusher	T.J. Watt	0.565
Severity BT	Rusher	Nick Bosa	0.524
Severity BT	Rusher	Myles Garrett	0.482
Severity BT	Rusher	Jaelan Phillips	0.441
Severity BT	Blocker	Halapoulivaati Vaitai	0.261
Severity BT	Blocker	Corey Linsley	0.261
Severity BT	Blocker	Tytus Howard	0.257
Severity BT	Blocker	Trent Williams	0.237
Severity BT	Blocker	Joe Thuney	0.230

5.3 Weekly Path Uncertainty

The weekly path bootstrap adds longitudinal uncertainty to the end-of-season summary. In this run we obtain 29 cumulative checkpoints for each player-model-role series. Figure 3 shows representative paths for high-impact rushers and blockers in each task.

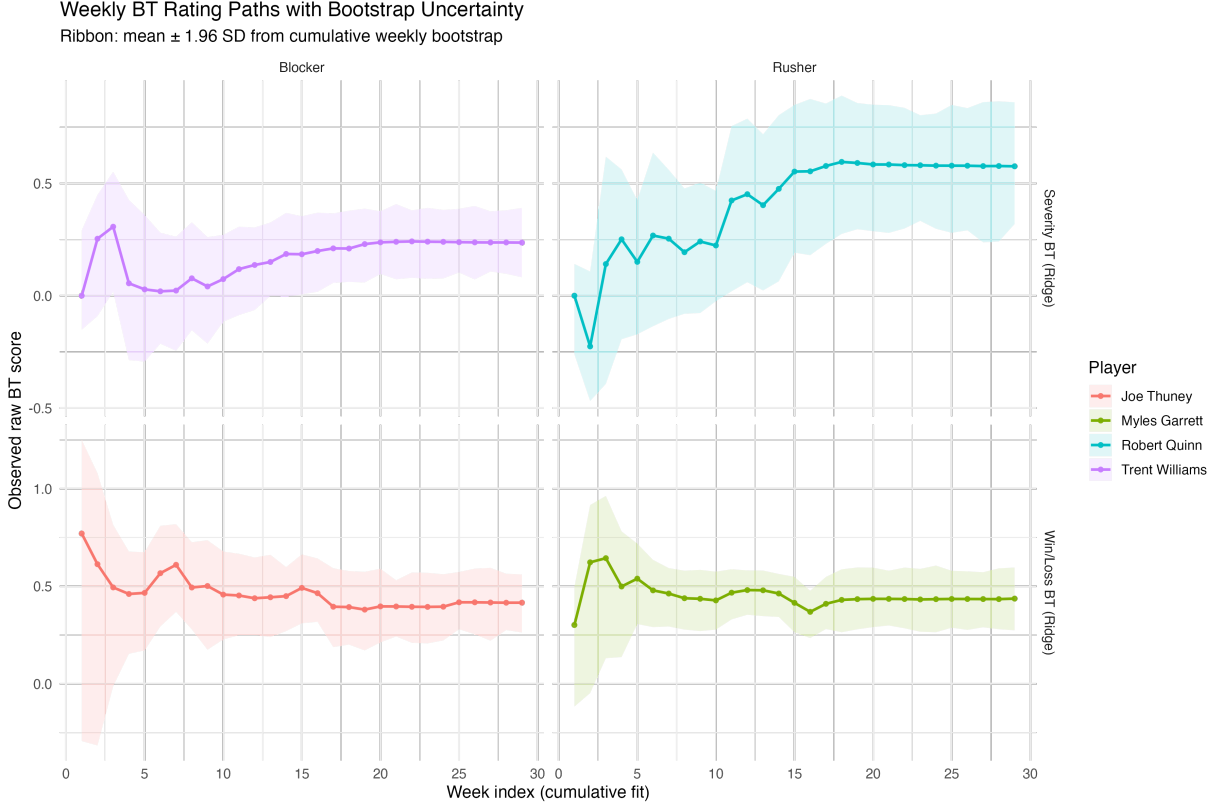


Figure 3: Weekly cumulative raw BT score paths with bootstrap uncertainty ribbons (mean $\pm$ 1.96 SD).

6 Discussion

The current BT pipeline provides a clearer statistical foundation than the previous pure Elo workflow for this dataset. In particular:

1. Both tasks are estimated and validated separately against task-appropriate baselines.
2. Regularization is tuned directly by out-of-sample deviance/log loss.
3. Uncertainty is reported both at endpoint (end-to-end bootstrap) and over time (weekly path bootstrap).

Empirically, win/loss BT and severity BT both beat their respective baselines out of sample, and bootstrap intervals support that these gains are not driven by a single split. This makes BT the right primary model family for both technical reporting and paper presentation.

6.1 Limitations and Future Work

Several limitations suggest clear extensions. First, the win labeling rule is still a simplified distance-based heuristic. More realistic labels could incorporate pocket geometry, rusher lane, and whether a pressure event materially constrains the quarterback. Second, the current

pipeline does not explicitly model chip blocks or tight-end/running-back help beyond the coarse double-team indicator. Third, both sacks and proximity-based pressure are affected by quarterback time-to-throw. Adding richer contextual features may help separate pass-rush quality from play design and quarterback behavior. Finally, although BT adjusts for opponent strength, it does not explicitly model teammate effects or position-family structure; hierarchical shrinkage by role and multi-season pooling are natural next steps.

6.2 Conclusion

Regularized BT models provide a stable and interpretable framework for NFL pass-rush and pass-protection evaluation. In this project state, BT outperforms baselines in both binary win/loss and outcome-severity tasks, with uncertainty quantified at both final-score and weekly-path levels.

References

- Bradley, R. A. and Terry, M. E. (1952). Rank analysis of incomplete block designs: I. the method of paired comparisons. *Biometrika*, 39(3/4):324–345.
- Burke, B. (2018). We created better pass-rusher and pass-blocker stats: How they work. ESPN Analytics. Explainer article on pass-rush and pass-block win rate metrics.
- Efron, B. and Tibshirani, R. J. (1994). *An Introduction to the Bootstrap*. Chapman & Hall/CRC, New York.
- Friedman, J., Hastie, T., and Tibshirani, R. (2010). Regularization paths for generalized linear models via coordinate descent. *Journal of Statistical Software*, 33(1):1–22.
- Glickman, M. E. and Jones, A. C. (2025). Models and rating systems for head-to-head competition. *Annual Review of Statistics and Its Application*, 12:1–31.
- Nguyen, Q., Yurko, R., and Matthews, G. J. (2023). Here comes the STRAIN: Analyzing defensive pass rush in american football with player tracking data. *The American Statistician*, 77(4):353–370.